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CHEMISTRY

1) Given below are two statements :
Statement I : Ferromagnetism is considered as an extreme form of paramagnetism.
Statement II : The number of unpaired electrons in a Cr^{2+} ion (Z = 24) is the same as that of a Nd^{3+} ion (Z = 60).
In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both Statement I and Statement II are true
(2) Both Statement I and Statement II are false
(3) Statement I is true but Statement II is false
(4) Statement I is false but Statement II is true

2) Which Pair have same density ?

(1) Ni & Cu
(2) Cu & Zn
(3) Zn & Cr
(4) Mn & Fe

3) Which of the following is the use of potassium permanganate?

(1) Bleaching of wool, cotton and silk fibers.
(2) decolourisation of oils.
(3) In analytical chemistry.
(4) All of these.

4) Read the following statements.
(i) Chemistry of actinoids is complex in comparison to chemistry of lanthanoids.
(ii) Ce^{4+} is very good reducing agent.
(iii) Eu^{2+} is a strong reducing agent.
(iv) Out of all lanthanides Ce,Pr,Nd,Dy and Ho shows +4 oxidation state.
Which of the following is the correct code for the statements above?

(1) TTFF
(2) TFTF
(3) FTFT
(4) FTTF

5) If $KMnO_4$ is reduced by oxalic acid in an acidic medium then oxidation number of Mn changes from

(1) 4 to 2

(2) 6 to 4

(3) +7 to +2

(4) 7 to 4

6) Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown-red vapours of

(1) CrO_3

(2) $CrCl_3$

(3) CrO_2Cl_2

(4) Cr_2O_3

7) Which of the following ion(s) is/are oxidising in nature?

(1) $V^{2+}\ (E^{\circ}_{M2+/M} = -1.18)$

(2) $Mn^{3+}\ \left(E^{\circ}_{M^3{}_{+}/M^{2+}} = +1.57\right)$

(3) $Cr^{2+}\ \left(E^{\circ}_{M^{2+}/M} = -0.91\right)$

(4) None

8) An alloy of transition metal containing a non transition metal as a constituent is

(1) invar

(2) bronze

(3) chrome steel

(4) stainless steel

9) Which copper halide does not exist?

(1) CuF_2

(2) $CuCl_2$

(3) Cu_2I_2

(4) CuI_2

10) Standard reduction potential (M^{+2}/M) is maximum negative for :-

(1) Ti

(2) V

(3) Cr

(4) Mn

11) Correct decreasing order of ΔaH

(1) 3d > 4d > 5d

(2) 5d > 4d > 3d

(3) 5d > 3d > 4d

(4) 3d > 5d > 4d

12) Write decreasing order of basic strength ?

(1) $Na(OH)_3 > Yb(OH)_3 > Eu(OH)_3 > Tm(OH)_3$

(2) $Ce(OH)_3 > Sm(OH)_3 > Gd(OH)_3 > Tb(OH)_3$

(3) $Yb(OH)_3 > Lu(OH)_3 > La(OH)_3 > Sm(OH)_3$

(4) $Sm(OH)_3 > Gd(OH)_3 > Eu(OH)_3 > Yb(OH)_3$

13) ***Assertion (A) :-*** Br^- ions do not interfere in chromyl chloride test for Cl^- ***Reason (R) :-*** Br^- on oxidation with acidified $K_2Cr_2O_7$ liberates Br_2 which dissolves in NaOH to give a colourless solution.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is incorrect but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

14) $Cr_2O_7^{-2} \underset{y}{\overset{x}{\rightleftharpoons}} 2CrO_4^{-2}$

X and Y are respectively :

(1) X = OH^- Y = H^+

(2) X = H^+ Y = OH^-

(3) X = OH^- Y = H_2O_2

(4) X = H_2O_2 Y = OH^-

15) Highest stable Mn fluoride is MnF_4 where as highest Mn oxide is Mn_2O_7 due to –

(1) O atom is smaller the F.

(2) Oxygen has ability to form multiple bonds

(3) Mn^{7+} does not exist

(4) F can not-stabilise higher oxidation states.

16) Which of the following statements is false :-

(1) $Cr_2O_7^{2-}$ has a Cr – O – Cr bond

(2) CrO_4^{2-} is tetrahedral in shape

(3) $Na_2Cr_2O_7$ is a primary standard in volumetry

(4) $Na_2Cr_2O_7$ is more soluble than $K_2Cr_2O_7$

17) The ionic radius of Cr is minimum in which of the following compounds ?

(1) CrF_3

(2) $CrCl_3$

(3) Cr_2O_3

(4) K_2CrO_4

18) Total number of lanthanoids (4f-elements) having 5d electrons in their electronic configuration are :

(1) 3

(2) 4

(3) 5

(4) 6

19) Which of the following is not correct.

(1) Mixed oxides of lanthanoids are employed as catalyst in petroleum cracking.

(2) A good deal of misch metal is used in Al-based alloy to produce bullets shell and lighter flant.

(3) Selenium and tellurium are also found as metal selenides and tellurides in sulphide ores.

(4) The stability of +6 oxidation state decreases down the group in group 16.

20) Which of the following metals will not form an amalgam ?

(1) Gold

(2) Silver

(3) Zinc

(4) Iron

21) In BF_3, the B–F bond length is 1.30 Å, when BF_3 is allowed to be treated with Me_3N, it forms an adduct, $Me_3N \rightarrow BF_3$, The bond length of B–F in the adduct is :

(1) Greater than 1.30Å

(2) Smaller than 1.30 Å

(3) Equal to 1.30 Å

(4) None of these

22) Which of the following statements is true on the basis of back bonding ?

(1) $CHCl_3$ is more acidic than CHF_3

(2) Dimethyl ether acts as a better Lewis base but not disilyl ether (SiH_3–O–SiH_3)

(3) $(CH_3)_3C$–O–H is less acidic than $(CH_3)_3Si$–O–H

(4) All of the above

23) In diborane, the two H_t–B–H_t and H_b–B–H_b angles are nearly :-

(1) 60°, 120°

(2) 120°, 97°

(3) 95°, 150°

(4) 120°, 180°

24) Which dimerisation reaction shows conversion of central atom from e^- deficient to e^- sufficient :-

(1) $2BH_3 \rightarrow B_2H_6$

(2) $2BeH_2 \rightarrow Be_2H_4$

(3) $2AlCl_3 \rightarrow Al_2Cl_6$

(4) All

25) Which of following occurs as a consequence of inert pair effect :-
(a) $SnCl_2$ act as a reducing agent
(b) $SnCl_4$ act as an oxidising agent
(c) SnO_2 is amphoteric
(d) CCl_2 is unstable but $PbCl_2$ is stable.
(e) PbO_2 is a good oxidising agent.

(1) a, d, e

(2) d, e

(3) a, b, c, d, e

(4) a, b, c

26) **Assertion :** PbI_4, BiI_5, FeI_3, CuI_2 are not exist.
Reason : Higher oxidation state of metal ions are strong oxidising agent which give redox reaction with Iodide ion.

(1) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are correct but **Reason** is NOT correct for **Assertion**.

(3) **Assertion** is correct but **Reason** is incorrect.

(4) **Assertion** is incorrect but **Reason** is correct.

27) Which one of the statement about H_3BO_3 is not correct :-

(1) It is weak monobasic acid

(2) It is lewis acid

(3) It do not act as proton donor

(4) It is a strong tribasic acid

28) The true statement for the acids of phosphorus, H_3PO_2, H_3PO_3 and H_3PO_4 is :

(1) The order of their acidity is :
$H_3PO_4 > H_3PO_3 > H_3PO_2$.

(2) All of them are reducing in nature.

(3) All of them are tribasic acids

(4) The geometry of phosphorus is tetrahedral in all the three

29) Peroxo bond (–O–O–) is present in
(i) Na_2O_2 (ii) CrO_5
(iii) PbO_2 (iv) SrO_2

(1) (i, ii, iv)

(2) (i, ii, iii, iv)

(3) (i, ii, iii)

(4) (ii, iii, iv)

30) Match **List-I** with **List-II** :

	List-I (Oxoacids of Sulphur)		**List-II (Bonds)**
A.	Peroxodisul-phuric acid	I.	Two S-OH, Four S=O, One S-O-S
B.	Sulphuric acid	II.	Two S-OH, One S=O
C.	Pyrosulphuric acid	III.	Two S-OH, Four S=O, One S-O-O-S
D.	Sulphurous acid	IV.	Two S-OH, Two S=O

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I, D-II

(2) A-I, B-III, C-IV, D-II

(3) A-III, B-IV, C-II, D-I

(4) A-I, B-III, C-II, D-IV

31) Which one of the following reactions does not come under hydrolysis type reaction ?

(1) $Li_3N + H_2O \rightarrow Li(OH) + NH_3$

(2) $P_2O_5 + H_2O \rightarrow H_3PO_4$

(3) $SiCl_4 + H_2O \rightarrow SiO_2 + HCl$

(4) $Cl_2 + H_2O \rightarrow HOCl + HCl$

32) Which of the following halides does not hydrolysed at room temperature ?

(1) $PbCl_4$

(2) $SiCl_4$

(3) CCl_4

(4) $SnCl_4$

33) Which of the following is not correctly matched ?

(1) Mg_2C_3 - Allylide

(2) CaC_2 - Acetylide

(3) Be_2C - Methanide

(4) Al_4C_3 - Allylide

34) Which of the following is not obtained by the hydrolysis of XeF_2 ?

(1) Xe
(2) HF
(3) XeO_3
(4) O_2

35)

Incorrect statement among the following is :

(1) Graphite has layered structure and layers are held by Vander Waal's forces.
(2) Density of diamond is more than that of graphite.
(3) C_{60} contain 12 hexagonal rings and 20 pentagonal rings.
(4) Graphite's cleaves easily between layers so used as a dry lubricant.

36) $\text{Red P} \xrightarrow[\text{sealed tube}]{803K} X$

$\text{White P} \xrightarrow[\substack{\text{Inert atmosphere} \\ \text{(for many days)}}]{573K} Y$

here X and Y are resp.

(1) α, and β black P
(2) β black P and Red P
(3) α Black P and Red P
(4) White P and Red P

37) Which of the following is dry and permanent bleaching agent :-

(1) O_3
(2) Moist SO_2
(3) H_2O_2
(4) Cl_2 water

38) Consider the following chemical reaction

$\underset{\text{(Formic acid)}}{HCOOH(aq.)} \xrightarrow{X,\ \Delta} CO(g) + H_2O$

In the above reaction (X) is –

(1) Conc. H_2SO_4
(2) Conc. HCl
(3) Conc. HNO_3
(4) Dil. HNO_3

39) In borax $Na_2BxOy(OH)z.WH_2O$ then x + y + z + w is.

(1) 17

(2) 21

(3) 23

(4) 24

40) Hybrid state of Aluminium in acidified aqueons solution of $AlCl_3$ is :-

(1) sp^3

(2) sp^3d

(3) sp^3d^2

(4) sp^3d^3

41) Match the column :-

	Column-I		**Column-II**
(a)	SiO_2	(i)	Acidic
(b)	GeO_2	(ii)	Basic
(c)	SnO_2	(iii)	Neutral
(d)	PbO_2	(iv)	Amphoteric

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(i), b-(i), c-(iv), d-(iv)

(3) a-(i), b-(iii), c-(iv), d-(iv)

(4) a-(i), b-(ii), c-(iv), d-(iii)

42)

Match **List-I** with **List-II :**

	List-I (Atom/Molecules)		**List-II (Property)**
(A)	Nitrogen atom	I.	Paramagnetic
(B)	Fluorine molecule	II.	Most reactive element in group 18
(C)	Oxygen molecule	III.	Element with highest ionisation enthalpy in group 15
(D)	Xenon atom	IV.	Strongest oxidising agent

Identify the **correct** answer from the options given below :

(1) A-III, B-I, C-IV, D-II

(2) A-I, B-IV, C-III, D-II

(3) A-II, B-IV, C-I, D-III

(4) A-III, B-IV, C-I, D-II

43) Identify the correct statements regarding 16 group elements.

(I) Reducing property of dioxide increases from SO_2 to TeO_2.

(II) Acidic nature decreases from SO_3 to TeO_3.
(III) Stability of halides decreases in the order
$F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$
(IV) Amongst tetrafluorides SF_4 a gas and TeF_4 solid.
Choose the correct answer from the option given below :

(1) I, II and III only
(2) I, II, III and IV
(3) II and IV only
(4) I, II and IV only

44) Which one is not present in the atmosphere ?

(1) Ar
(2) Xe
(3) He
(4) Rn

45) Which has the lowest boiling point of any known substance?

(1) H_2
(2) He
(3) N_2
(4) O_2

ANSWER KEYS

CHEMISTRY

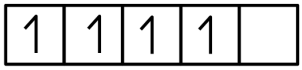
Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	4	2	3	3	2	2	4	1	2	2	4	1	2	3	4	1	2	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	4	2	3	1	1	4	4	1	1	4	3	4	3	3	3	1	1	2	3
Q.	41	42	43	44	45															
A.	2	4	3	4	2															

SOLUTIONS

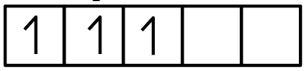
CHEMISTRY

1)

Ferromagnetism > Paramagnetism
Statement-I is true

$Cr^{+2} = 3d^4$ 

4 unpaired electrons

$Nd^{+3} = 4f^3$ 

3 unpaired electrons

Statement-II is false.

2)

NCERT Pg. # 220

3)

NCERT Pg. # 239

4)

NCERT Pg. # 234-238

5)

$KMnO_4 + H_2SO_4 + H_2C_2O_4 \rightarrow MnSO_4 + K_2SO_4 + CO_2 + H_2O$

6)

$NaCl + K_2Cr_2O_7 + H_2SO_4 \rightarrow CrO_2Cl_2$ (Red orange vapour) chromyl chloride

7)

Positive E^0 value more favourable process.

8)

Invar → Ni(36%), Bronze (Cu + Sn)
Chrome steel ⇒ Cr(2-4%) Stainless steel Cr(12-14%), Ni(2-4%)

9)

Explanation - This question tests knowledge of copper halide stability, specifically why CuI_2 is unstable and decomposes into CuI and iodine.

Concept -(Copper halide stability all cases)
The copper halide that does not exist is CuI_2. Here's why:

A. **Cu(II) and Iodide:**
   A. Cu^{2+} ions are unstable in the presence of I^- ions.
   B. The I^- ion is a strong reducing agent.
   C. It readily reduces Cu^{2+} to Cu^+.
   D. The reaction that occurs is: $2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$.
B. **Result:**
   A. Instead of forming CuI_2, copper(I) iodide (CuI) is formed, along with iodine (I_2).

Therefore the answer is 4. CuI_2.

Final Answer - Option (4)

10) NCERT-XII, table (8.2) (old)

11) NCERT-XII, Pg. # 219 (old)

12)

Left to right basic strength decreases (f-block elements)

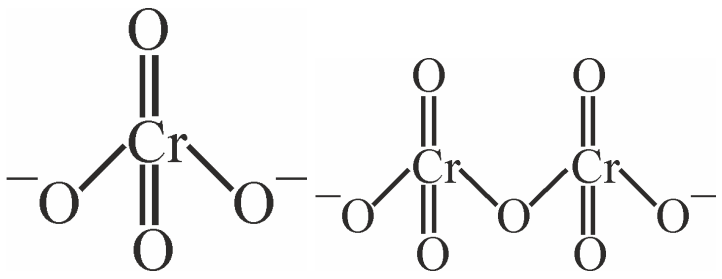
13) Br^- does not interfere with the chromyl chloride test, because Br^- converted into Br_2 and liberated which leaves NaOH solution colourless.

14) $Cr_2O_7^{-2} \underset{H^+}{\overset{OH^-}{\rightleftharpoons}} 2CrO_4^{-2}$

15)

NCERT Pg. # 226

16)



$Na_2Cr_2O_7$ is not used as a primary standared in volumatry due to its moisture absorbing nature

17)

Positive charge $\propto \frac{1}{\text{Ionic radius}}$

18) Atomic number: 58, 64, 71

19) **Question Explanation:** Identify incorrect statement
Concept: This question based on Alloys (Misch metal)
Solution: Misch metals significant use in Al – based bullet shells incorrect; It's primarily for lighter flints.
Final Answer: (2)

20) **Question Explanation**: Identifying the metal that will not form an amalgam.
Concept: Amalgam Formation.
Solution: Iron(Fe),Cobalt(Co),and Nickel(Ni) are exceptions and do not form amalgams readily with Hg. **Final Answer**: Iron.

21) **Explain Question:** Compare B–F bond length in BF_3 and its adduct with Me_3N.
Concept: π-back bonding
Solution: In BF_3, strong back-bonding shortens the bond. Adduct formation uses Boron's empty orbital, stopping back-bonding. The B–F bond character changes from partial double to single, hence bond length increases.
Final Answer: Option (1)

22)

Asking for: Identifying the true statement based on the concept of back bonding.

Concept: Back Bonding (pπ-dπ), Acidity, Lewis Basicity, Influence of d-orbitals.

Solution:

A. **$CHCl_3$ is more acidic than CHF_3:** The acidity of these compounds depends on the stability of the conjugate base (CX_3^-). While CCl_3^- can have some stabilization through pπ-dπ back bonding between the lone pair on carbon and the vacant d-orbitals of chlorine.Thus, this statement is true.

B. **Dimethyl ether acts as a better Lewis base but not disilyl ether (SiH_3OSiH_3):** In disilyl ether, the oxygen lone pairs can participate in pπ-dπ back bonding with the vacant d-orbitals of silicon. This delocalization reduces the availability of the lone pairs for donation to a Lewis acid, making it a weaker Lewis base than dimethyl ether, where such back bonding is not possible with carbon. Thus, this statement is **true** due to back bonding.

C. **$(CH_3)_3COH$ is less acidic than $(CH_3)_3SiOH$:** The acidity of these compounds depends on the stability of the conjugate base (alkoxide/siloxide). In $(CH_3)_3SiO^-$, the negative charge on oxygen can be stabilized bypπ-dπ back bonding with the vacant d-orbitals of silicon. This stabilization makes $(CH_3)_3SiOH$ more acidic than $(CH_3)_3COH$, where such stabilization is absent. Thus, this

statement is **true** due to back bonding.

D. **All of the above:** Since all of the above statements are correct.

Answer : Option : 4

23)

Terminal bond angle > Bridging bond angle

24)

Asking for: Identifying the dimerization reaction(s) where the central atom (B, Be, Al) changes from electron-deficient to electron-sufficient.

Concept: Electron Deficiency, Dimerization Reactions, Bridging Bonds, Lewis Acidity, Octet Rule.

Solution:

A. **$2BH_3 \rightarrow B_2H_6$ (Diborane):**

   A. In BH_3, Boron (B) has 3 valence electrons, forming 3 bonds with Hydrogen (H), resulting in 6 electrons around B. Boron is electron-deficient.
   B. In B_2H_6, each Boron atom is bonded to two terminal H atoms and shares two bridging H atoms. These bridging bonds are 3-center-2-electron bonds and does not contribute any extra electron. So B_2H_6 is still an electron-deficient molecule.
   C. **Conversion: Electron-deficient B → Electron-deficient B.**

B. **$2BeH_2 \rightarrow Be_2H_4$ (Beryllium Hydride Dimer):**

   A. In BeH_2, Beryllium (Be) has 2 valence electrons, forming 2 bonds with H, resulting in 4 electrons around Be. Beryllium is electron-deficient.
   B. In Be2H4, each Be atom is bonded to one terminal H and shares two bridging H atoms. Similar to diborane, these bridging bonds do not contribute any extra electron and thus Be2H4 molecule remains electron-deficient .
   C. **Conversion: Electron-deficient Be → Electron-deficient Be.**

C. **$2AlCl_3 \rightarrow Al_2Cl_6$ (Aluminum Chloride Dimer):**

   A. In $AlCl_3$, Aluminum (Al) has 3 valence electrons, forming 3 bonds with Chlorine (Cl), resulting in 6 electrons around Al. Aluminum is electron-deficient.
   B. In Al_2Cl_6, each Al atom is bonded to two terminal Cl atoms and two bridging Cl atoms. The bridging involves coordinate covalent bonds where Chlorine donates a lone pair to the electron-deficient Aluminum. Through these bridging bonds, each Al effectively achieves an octet.
   C. **Conversion: Electron-deficient Al → Electron-sufficient Al.**

Answer : Option : 3

25) **Concept:**

inert pair effect

Solution:
• $SnCl_2$ act as reducing agent because $Sn^{+4} > Sn^{+2}$
• $PbCl_2$ stable because $Pb^{+2} > Pb^{+4}$
• PbO_2 is good oxidising agent because $Pb^{+2} > Pb^{+4}$.

26) **Explanation :**
The question provides an assertion that certain compounds (PbI_4, BiI_5, FeI_3, CuI_2) do not exist and a reason stating that higher oxidation states of metal ions are strong oxidizing agents that undergo redox reactions with iodide ions. You need to determine if both the assertion and the reason are correct, and if the reason accurately explains why these specific compounds do not exist.

Concept :
Inert Pair Effect

Solution :
Assertion : PbI_4, BiI_5, FeI_3, CuI_2 do not exist due to instability.
Reason : Higher oxidation stats of metal ions act as strong oxidizing agents, reacting with iodide ions.
The instability of FeI_3 and CuI_2 is directly due to the redox reaction with iodide, as stated in the reason. For PbI_4 and BiI_4 while the inert pair effect is primary, the reducing nature of iodide also contributes to their non-existence in higher oxidation states. Thus, the reason is a correct explanation of Assertion.

Final Answer :
Option (1)

27) **Explanation :**
Let's analyze the given statements about H_2BO_3 (Boric Acid) and determine which one is incorrect:

Concept : (Hydrolysis of H_3BO_3)
Boric acid is a weak monobasic acid. It does not act as a proton donor in the usual sense. Instead, it accepts a hydroxide ion (OH^-) from water, releasing a proton (H^+) into the solution. This makes it behave like a Lewis acid (electron pair acceptor). The reaction can be represented as:

$H_3BO_3 + H_2O \rightleftharpoons [B(OH)_4]^- + H^+$

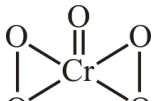
So, while it increases the H^+ concentration, it does so by inieracting with water rather than directly donating a proton itself. This is why it's considered a Lewis acid and a weak monobasic acid.

Final Answer : Option(4)

28)

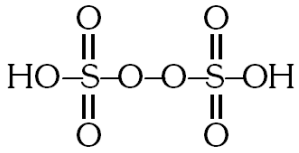
Concept :- Oxyacid's of Phosphorus
Solution :-
(1) Correct order of acids eq $H_3PO_4 < H_3PO_3 < H_3PO_2$
(2) $H_3PO_4 \rightarrow$ Oxidising agent
$H_3PO_3 \rightarrow$ Can act as both oxidising and reducing agent
$H_3PO_2 \rightarrow$ Reducing agent
(3) H_3PO_2 is monobasic oxide
(4) This is these statement as in all oxyacids phosphorus is tetrahedral in nature.
Answer :- 4
The geometry of phosphorus is tetrahedral in all the three

29) (i) $Na_2O_2 \rightarrow 2Na^+ + O_2^{-2}$ peroxo

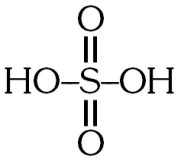


(ii) O O here 2 peroxo linkage
(iii) PbO_2 is a normal oxide
(iv) $SrO_2 \rightarrow Sr^{+2} + O_2^{-2}$ peroxo

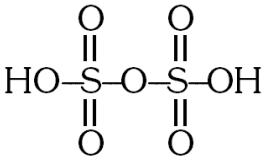
30) A → Peroxodisulphuric acid



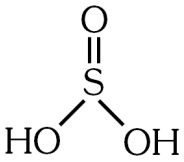
B → Sulphuric acid
H_2SO_4



C → Pyrosulphuric acid $H_2S_2O_7$



D → Sulphurous acid H_2SO_3



31)

Asking for : The reaction among the given options that is not a hydrolysis reaction.

Solution :

The reaction of chlorine with water ($Cl_2+H_2O \rightleftharpoons HOCl+HCl$) stands out as potentially not fitting

the strict definition of hydrolysis. Hydrolysis typically involves the cleavage of a molecule by water, with the H^+ and OH^- fragments becoming bonded to the resulting parts.

The other options (Li_3N, P_2O_5, $SiCl_4$) clearly demonstrate hydrolytic cleavage where water directly attacks and breaks bonds, with its components becoming integral parts of the products.

Answer : Option : 4

32) **Explain Question:** Identify the halide that **does NOT hydrolyse** at room temperature.

Concept: Hydrolysis & Covalent Character.

Solution: CCl_4 is highly covalent and the central carbon atom has no empty d-orbitals for water to attack or expand its coordination number. Also, C–Cl bonds are very strong and carbon is too small to be attacked by water. Therefore, CCl_4 resists hydrolysis, while $SiCl_4$, $SnCl_4$, $PbCl_4$ have vacant d-orbitals and larger size, allowing easy hydrolysis.

Final Answer: Option (3)

33) **Question Explanation:**
The question asks us to identify the metal carbide and its corresponding anion name that are incorrectly matched.

Concept: Anion name formed by metal carbide)

A. $Be_2C + 2H_2O \rightarrow 2Be(OH)_2 + CH_4$
B. $Al_4C_3 + 12H_2O \rightarrow 4Al(OH)_3 + 3CH_4$
C. $CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$
D. $Mg_2C_3 + 4H_2O \rightarrow 2Mg(OH)_2 + C_3H_4$

Therefore, the correct answer is: Option (3). CaC_2
Answer – Option (4)

34) **Question Explanation:** Find the product not formed by XeF2 hydrolysis.
Concept: Hydrolysis of noble gas compounds.
Solution
The hydrolysis of xenon difluoride (XeF_2) is a redox reaction. When XeF_2 reacts with water, xenon is reduced from a +2 oxidation state to 0, and oxygen from water is oxidized from -2 to 0. The balanced chemical reaction for the hydrolysis is: $2XeF_2(s)+2H_2O(l)\rightarrow 2Xe(g)+4HF(aq)+O_2(g)$ The products of this reaction are xenon (Xe), hydrogen fluoride (HF), and oxygen (O_2). Xenon trioxide (XeO_3) is a product of the hydrolysis of XeF_4 and XeF_6, but not XeF_2.
Final Answer:(3) XeO_3

35)

Explain Question: Find the incorrect statement about carbon allotropes.
Concept: Structure of fullerene

Solution: C_{60} actually contains **12 pentagonal** and **20 hexagonal** rings, not the reverse. Hence, the given statement about ring count is incorrect.
Final Answer: Option (3)

36) **Explanation:**
In this question reaction of Red and white phosphorus is asked.
Concept:
Reaction of red and white phosporus is used.
Solution:

Red P $\xrightarrow[\text{sealed tube}]{803\ K}$ α-Black

White P $\xrightarrow[\text{Inert atmosphere}]{573\ K}$ Red P

37) **Explanation :**
The question asks you to identify the dry and permanent bleaching agent from the given options.

Concept : (Mechanism of bleaching action)
The dry and permanent bleaching agent among the options is 1.O_3 (Ozone).

A. **Ozone (O_3):** Ozone is a powerful oxidizing agent and bleaches materials without leaving any residue. It's considered a dry bleaching agent because it doesn't involve water in its bleaching action. The bleaching effect is also permanent as it chemically alters the color-causing substances.
B. **Moist SO_2 (Sulfur dioxide):** SO_2 bleaches by reduction, which is often temporary. It's also not a dry process as it requires moisture. Colors bleached by SO_2 can often return over time.
C. **H_2O_2 (Hydrogen peroxide):** While H_2O_2 is a good bleaching agent. it's not considered a dry bleaching agent. It's typically used in aqueous solutions.
D. **Cl_2 water (Chlorine water);** Chlorine water is a strong bleaching agent, but it's not dry and can damage some materials. Like H_2O_2, it involves water in its bleaching process.

Final Answer option (1)

38)

Conc. H_2SO_4 act as dehydrating agent.

39)

Explanation – The question is asking for the sum of the values of x + y + z + w in the formula of borax, which is given as $Na_2B_xO_y(OH)_z \cdot W\ H_2O$.

Concept – (Chemical formula of hydroxides)

Borax is $Na_2B_4O_7 \cdot 10H_2O$. Therefore:

A. x = 4 (boron atoms)

B. y = 7 (oxygen atoms)

C. z = 0 (no OH groups explicitly shown in this simplified formula - the water of hydration accounts for the H and O)

D. w = 10 (water molecules)

x + y + z + w = 4 + 7 + 0 + 10 = 21

So the correct answer is 2. 21.

40) $AlCl_3 + 6H_2O \rightarrow \underset{sp^3d^2}{[Al(H_2O)_6]^{+3}_{(aq)}} + 3\ Cl^-(eq)$

41) **Explanation :** This question asks you to match the oxides of SiO_2, GeO_2, SnO_2, and PbO_2 with their corresponding chemical properties (acidic, basic, natural, or amphoteric). We need to understand the general behavior of these oxides and their interaction with acids and bases.
Concept : (Chemical properties of oxides of SiO_2, GeO_2, SnO_2, & PbO_2)
The correct matching is 2. a-(i), b-(i), c-(iv), d-(iv)
Here's why :
SiO_2 (Silicon Dioxide) : Acidic Silicon dioxide is the acidic anhydride of silicic acid. It reacts with bases to form silicates.
GeO_2 (Germanium Dioxide) : Acidic Similar to SiO_2, germanium dioxide is also acidic, through slightly less so.
SnO_2 (Tin Dioxide) : Amphoteric Tin dioxide can react with both acids and bases, exhibiting amphoteric behavior.
PbO_2 (Lead Dioxide) : Amphoteric Lead dioxide also exhibits amphoteric properties. It reacts with strong acids to form lead (IV) salts and with strong bases to form plumbates.
Final Answer : (2)

42)

(A)	Nitrogen atom	Element with highest IP in group 15
(B)	Fluorine molecule	Strongest oxidising agent
(C)	Oxygen molecule	Paramagnetic (MOT)
(D)	Xenon atom	Most reactive element in group 18

43) NCERT-XII, Pg.No. 188

44) NCERT-XII, Pg. # 208 (old)

45) NCERT Pg. No. # 203